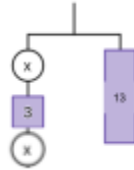




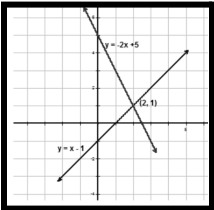
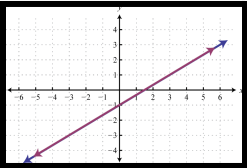
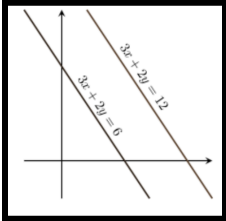
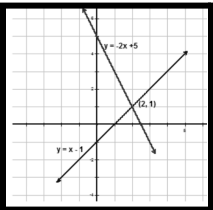
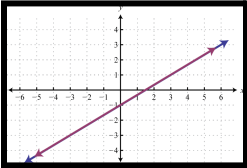
Name _____ HR _____

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8th Math Unit 4 - System of Equations STAR Cards!

QUESTION	ANSWER																								
4a) ☆☆☆ What does it mean to <u>solve</u> an equation? Example: $2x + 3 = 13$ 	4a) <i>Solving</i> an equation means finding the value of the variable that will make the left side equal the right side. Example: $x = 5$																								
4b) ☆☆☆ Most equations have <u>one solution</u>, but some equations have no solutions, or more than one solution. Mr. Cox solved three different equations and each of his last lines of work are below. Identify: 1) the <u>number</u> of solution(s) 2) <u>explain why</u> <table><tr><th>Mr. Cox got:</th><th># of solutions</th><th>Explain</th></tr><tr><td>$x = 0$</td><td></td><td></td></tr><tr><td>$5 = -5$</td><td></td><td></td></tr><tr><td>$9 = 9$</td><td></td><td></td></tr></table>	Mr. Cox got:	# of solutions	Explain	$x = 0$			$5 = -5$			$9 = 9$			4b) <table><tr><th>Mr. Cox got:</th><th># of solutions</th><th>Explain</th></tr><tr><td>$x = 0$</td><td>ONE SOLUTION</td><td>because x equals only one number</td></tr><tr><td>$5 = -5$</td><td>NO SOLUTION</td><td>Because it's never true. 5 does not = -5.</td></tr><tr><td>$9 = 9$</td><td>INFINITE SOLUTIONS</td><td>because it's always true. 9 always equals 9.</td></tr></table>	Mr. Cox got:	# of solutions	Explain	$x = 0$	ONE SOLUTION	because x equals only one number	$5 = -5$	NO SOLUTION	Because it's never true. 5 does not = -5.	$9 = 9$	INFINITE SOLUTIONS	because it's always true. 9 always equals 9.
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QUESTION	ANSWER
4c)  Why can't we find Mr. Cox and Ms. Bank's age with this equation: $C + B = 55$ C = Cox's Age B= Banks' Age	4c) We can't solve this because we have two different variables in only one equation.
4d)  What is a <u>system of equations</u>?	4d) Two related equations that have the same variables and must be solved together Ex: $C + B = 55$ $B = 2C - 41$
4e)  What is a good first step to solve this system of equations? Explain. $C + B = 55$ $B = 2C - 41$	4e) • Since the 2nd equation says, B equals $2C - 41$ • You can replace the "B" in the first equation with "$2C - 41$" Now you can solve: $C + \underline{2C - 41} = 55$
4f)  For the system below: $y + 2x = 5$ $y = 4x - 7$ The solution is $x = 2$ and $y = 1$ 1) How would you express this answer as a coordinate pair? 2) For this system of equations, what would be happening at this coordinate pair?	4f) 1) $(2,1)$ 2) The two lines would intersect at the point $(2,1)$ on the graph

QUESTION	ANSWER
4g) These graphs represent 3 separate systems of equations. Identify the number of solutions each graph has. 1  2  3 	4g) 1) ONE SOLUTION  2) INFINITE SOLUTIONS  3) NO SOLUTIONS 